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ONE HUNDRED AND TWO PROBLEMS IN MATHEMATICAL LOGIC

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This expository paper contains a list of 102 problems which, at the time of publication, are unsolved. These problems are distributed in four subdivisions of logic: model theory, proof theory and intuitionism, recursion theory, and set theory. They are written in the form of statements which we believe to be at least as likely as their negations. These should not be viewed as conjectures since, in some cases, we had no opinion as to which way the problem would go.

In each case where we believe a problem did not originate with us, we made an effort to pinpoint a source. Often this was a difficult matter, based on subjective judgments. When we were unable to pinpoint a source, we left a question mark. No inference should be drawn concerning the beliefs of the originator of a problem as to which way it will go (lest the originator be us).

The choice of these problems was based on five criteria. Firstly, we are only including problems which call for the truth value of a particular mathematical statement. A second criterion is the extent to which the concepts involved in the statements are concepts that are well known, well defined, and well understood, as well as having been extensively considered in the literature. A third criterion is the extent to which these problems have natural, simple and attractive formulations. A fourth criterion is the extent to which there is evidence that a real difficulty exists in finding a solution. Lastly and unavoidably, the extent to which these problems are connected with the author's research interests in mathematical logic.

Each problem is followed by notes indicating, in each case, the principal partial results, and the principal modifications of the problem that remain unsolved (as far as we were able to ascertain).

Due to the criteria used in selecting these problems, several important research projects, which are significant for the development of mathematical logic and which can be expected to yield additional problems that meet the criteria used in our selection, are not touched upon here. We plan to report on these projects in a future expository paper.

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MODEL THEORY

Automorphisms.

1. *If a theory in $\mathcal{L}_{\omega\omega}$ with finitely many nonlogical symbols has a rigid model of power κ , it has rigid models of each uncountable power $< \kappa$ (Ehrenfeucht)*

2. *If a theory in $\mathcal{L}_{\omega\omega}$ with finitely many nonlogical symbols has rigid models of each power $< \omega_{\omega_1}$, it has rigid models of each uncountable power (Ehrenfeucht).*

The statement is false if ω_{ω_1} is replaced by any ω_α , $\alpha < \omega_1$ (see [Eh]).

Axiomatic model theory.

3. *No logic strictly between $\mathcal{L}_{\omega\omega}$ and $\mathcal{L}_{\infty\infty}$ obeys the interpolation theorem.*

$\mathcal{L}_{\omega\omega}$ and $\mathcal{L}_{\infty\infty}$ do not obey the interpolation theorem (see [Mal]). The statement is open if “interpolation” is replaced by “definability”.

4. *There is a logic satisfying the full compactness theorem as well as the interpolation theorem, which is not equivalent to $\mathcal{L}_{\omega\omega}$ even for finite models.*

See [Sh] for terminology. Every logic satisfying the countable compactness theorem and the downward Lowenheim-Skolem Theorem is equivalent to predicate calculus (see [Li] and [Fr 4]). There is a logic satisfying full compactness, which is not equivalent to predicate calculus for countable models (see [Sh]). It is open whether every logic satisfying the countable compactness theorem and the interpolation theorem is equivalent to predicate calculus.

Beth’s definability theorem.

5. For logics $\mathcal{L}$, $\Delta(\mathcal{L})$ is the logic whose elementary classes consist of those PC classes in $\mathcal{L}$ whose complements are also PC in $\mathcal{L}$. *The logic $\Delta(\mathcal{L}_{\omega\omega})$ does not satisfy the definability theorem.*

$\mathcal{L}_{\omega\omega}$ does not satisfy the definability theorem (see [Gr]). $\Delta(\mathcal{L}_{\omega\omega})$ does not satisfy the interpolation theorem (see [Fr 7] and [Hut]). It is open whether $\Delta(\mathcal{L}_{\infty\infty})$ obeys the definability theorem, or even the interpolation theorem.

6. Q_α is the quantifier “there exists at least ω_α many”. *The logic $\Delta(\mathcal{L}(Q_1))$ does not satisfy the definability theorem.*

$\Delta(\mathcal{L}(Q_1))$ does not satisfy the interpolation theorem (see [Fr 7] and [Hut]).

7. The weak definability theorem differs from the definability theorem in that “every structure has at most one expansion such that $\dots$ ” is replaced by “every structure has exactly one expansion such that $\dots$ ” in the hypothesis. *The weak definability theorem fails in $\mathcal{L}_{\kappa\omega}$, for $\omega_1 < \kappa \leq \infty$.*

The definability theorem fails in $\mathcal{L}_{\kappa\omega}$, for $\omega_1 < \kappa \leq \infty$ (see [Gr]). The definability theorem, or even the interpolation theorem holds in $\mathcal{L}_{\omega_1\omega}$ (see [Lo]).

8. *The weak definability theorem fails in $\mathcal{L}(Q_1)$.*

The definability theorem fails in $\mathcal{L}(Q_1)$ (see [Fr 11]).

Hanf numbers.

9. *If κ is the first Ramsey cardinal, then the Hanf number of $\mathcal{L}_{\omega_1\omega_1}$ is greater than κ (Kunen).*

This statement, with “Ramsey” replaced by “measurable”, is consistent with and independent of ZFC + “there is a measurable cardinal” (see [Ku], [Mag 2])

respectively). The Hanf number of $\mathcal{L}_{\omega_1\omega_1}$ is greater than certain types of Ramsey cardinals (see [Si 1]).

10. *There is a transitive model A of ZF such that $A \cap On = \alpha < \omega_1$, $|A| = \beth_\alpha$, $\mathcal{L}_A$ obeys the Barwise compactness theorem, the definability theorem, and the Hanf number of $\mathcal{L}_A$ is $\beth_\alpha$.*

Gregory has shown that $ZFC +$ “there exists an uncountable admissible set A such that $A \cap On < \omega_1$ and $\mathcal{L}_A$ obeys the Barwise compactness theorem” is consistent. There is a transitive model A of ZF such that $A \cap On = \alpha < \omega_1$, A does not satisfy the definability theorem, and the Hanf number of $\mathcal{L}_A$ is $\beth_\alpha$ (see [Fr 14]). For every countable admissible set A with $A \cap On = \alpha$, $\mathcal{L}_A$ obeys interpolation, and the Hanf number of $\mathcal{L}_A$ is $\beth_\alpha$ (see [Ba]).

Models of analysis.

11. *For $\omega \leq \kappa \leq \lambda \leq 2^\omega$, there is a model of the true sentences of second-order arithmetic, with κ many integers, λ many sets of integers, and nonstandard integers.*

The statement is commonly known for $\lambda \leq \omega_1$, or for $\kappa = \lambda$. No other information is known.

12. *There is an ω -model of the true sentences of second-order arithmetic which is not properly included in any ω -model of the true sentences of second-order arithmetic other than the full model.*

13. *Every ω -model of second-order arithmetic properly includes an ω -model of second-order arithmetic of the same cardinality. If the first is uncountable, the second may also be chosen to be elementary equivalent.*

It is commonly known that there are countable ω -models of second-order arithmetic with no elementary equivalent proper submodel. Every countable ω -model of second-order arithmetic properly includes another (see [Fr 9]).

Models of arithmetic.

14. *If T is any consistent extension of first-order arithmetic, then the order types of nonstandard models of T are exactly the order types of nonstandard models of first-order arithmetic.*

Henkin has shown that this statement is true for countable order types.

15. *If T is any recursive theory containing first-order arithmetic, and $(A, +, x)$ is a model of first-order arithmetic, then some $(A, +, \cdot)$ is a model of T .*

We have proved this statement for countable models.

16. *Every model of the true sentences of first-order arithmetic which is the arithmetic part of a model of second-order arithmetic is the arithmetic part of a model of type theory.*

We have proved this statement for countable models.

Number of models.

17. *Every sentence in $\mathcal{L}_{\omega\omega}$ with uncountably many countable models (up to isomorphism) has 2^ω such (Vaught).*

18. *There is a sentence in $\mathcal{L}_{\omega_1\omega}$ with exactly ω_1 countable models (up to isomorphism) (?).*

Every sentence in $\mathcal{L}_{\omega_1\omega}$ with more than ω_1 countable models (up to isomorphism) has 2^ω such (see [Mo]).

19. *The number of models (up to isomorphism) of power κ , of a theory in $\mathcal{L}_{\omega\omega}$, is a nondecreasing function of those $\kappa > \omega$ (Morley).*

20. *There is no complete finitely axiomatized theory which is ω_1 -categorical (Vaught).*

21. *There is no sentence in $\mathcal{L}(Q_1)$ with exactly one uncountable model.*

Shelah has shown, under the axiom of constructibility, that if a countable theory in $\mathcal{L}(Q_1)$ has $< 2^{\omega_1}$ models of power ω_1 , it has a model of power ω_2 .

22. *There exists a finitely axiomatized undecidable theory in $\mathcal{L}_{\omega\omega}$ with countably many complete extensions (Hanf).*

There exists a finitely axiomatized essentially undecidable theory of any prescribed recursively enumerable degree of unsolvability (see [Ha]). It is not known whether every axiomatizable theory is isomorphic to a finitely axiomatized theory (see [Ha]).

23. *There is a set of natural numbers which is the set of cardinalities of finite models of a sentence in $\mathcal{L}_{\omega\omega}$, but whose complement is not (Scholz).*

This statement is equivalent to there being no decision procedure for testing whether a formula in the propositional calculus is a tautology, in polynomial time (see [Jon S]).

Propositional logic.

24. Define $P(\kappa)$ for infinite regular κ , as infinitary propositional calculus based on κ atoms, negation, and conjunctions of each length less than κ . $P(\kappa)$ obeys the interpolation theorem if and only if κ is strongly inaccessible, or the successor of a strong limit cardinal of cofinality ω .

$P(\kappa)$ obeys the interpolation theorem if κ is the successor of a strong limit cardinal of cofinality ω (see [Fr 2]). $P(\kappa)$ does not obey the interpolation theorem if κ is the successor of a cardinal of cofinality $> \omega$ (see [Fr 2]).

Specific models.

25. *The first-order theory of any nonabelian free group is decidable (Tarski).*

The first-order theory of every free abelian group is decidable (see [Sz]). It is not known whether all nonabelian free groups are elementary equivalent.

26. *The first-order theory of $(\mathbf{R}, <, +, \times, \exp)$ is decidable (Grzegorzcyk).* The first-order theory of $(\mathbf{R}, <, +, \times)$ is decidable (see [Tar]).

27. *The real number π is not first-order definable in $(\mathbf{R}, <, +, \times, \exp)$ (?).*

Since π is not algebraic, π is not first-order definable in $(\mathbf{R}, <, +, \times)$ (see [Tar]).

28. *For each $n \geq 0$, there is a function $f: \mathbf{R} \rightarrow \mathbf{R}$ of class C^n such that for all functions $g: \mathbf{R} \rightarrow \mathbf{R}$ of class C^{n+1} , $(\mathbf{R}, <, f)$ is not isomorphic to $(\mathbf{R}, <, g)$.*

We have observed that this is true for $n = 0$. We have also observed that if f is continuous and monotone, there is a g of class C^∞ such that $(\mathbf{R}, <, f)$ is isomorphic to $(\mathbf{R}, <, g)$.

Transfer theorems.

29. *It is not provable in ZFC that $(\omega_2, \omega) \rightarrow (\omega_2, \omega_1)$ (?).*

It is commonly known, using ultraproducts, that $(\omega_2, \omega) \rightarrow (\omega_2, \omega_1)$ if $2^\omega = \omega_1$. It is not provable in ZFC that $(\omega_1, \omega) \rightarrow (\omega_2, \omega_1)$ (see [Mi]).

30. $(\omega_\omega, \omega) \rightarrow (2^\omega, \omega)$ (Shelah).

This statement is obvious if $2^\omega < \omega_\omega$.

PROOF THEORY AND INTUITIONISM

First- and second-order arithmetic.

31. *The free variable system with 0, ', +, · based on the axioms for successor, the recursion equations for +, ·, and the rule of induction is decidable* (Myhill).

This system cannot be complete by the undecidability of the existence of solutions to diophantine equations in the natural numbers (see [Mat]). This system is equivalent to a natural free variable system without the rule of induction (see [She]). The universal theory of rings with unit is decidable (see [Sim]). In light of [She], this is relevant.

32. *Let T be any consistent set of finitary propositional formulae (using atoms p_n) closed under classical entailment. Then there is a sequence A_n of Σ_1^0 sentences of first-order arithmetic such that the propositional formulae which result in sentences provable in classical first-order arithmetic when p_n is replaced by A_n are precisely the propositional formulae in T (?).*

This statement is true if T is the set of all classical tautologies (see [Mos]).

33. Let PA be classical first-order arithmetic. *There are sentences A, B of PA such that $PA \vdash A \leftrightarrow$ "to every proof in PA of A there is a proof in PA of $\sim A$ with smaller gödel number," $PA \vdash B \leftrightarrow$ "to every proof in PA of B there is a proof in PA of $\sim B$ with smaller gödel number," and not $PA \vdash A \leftrightarrow B$ (Kreisel).*

If $PA \vdash A \leftrightarrow$ "there is no proof in PA of A ," then $PA \vdash A \leftrightarrow$ " PA is consistent" (see [Go]). Sentences A obeying the condition in the statement are called Rosser sentences, and for no Rosser sentence A is any of $PA \vdash A \leftrightarrow$ " PA is consistent," $PA \vdash A \leftrightarrow$ " PA is not consistent," $PA \vdash A$, $PA \vdash \sim A$ true (see [Ro]).

34. Assume PA is formalized precisely as in [K11]. *Suppose that $A(x)$ is a formula of PA such that, for some k , each $A(\bar{n})$ has a proof in PA with $< k$ steps. Then $PA \vdash A$ (Kreisel).*

This statement is true if PA is modified by using the relations $+(a, b, c)$, $\cdot(a, b, c)$ in favor of $a + b = c$, $a \cdot b = c$, and modifying the axioms for $+$, $\cdot$ accordingly (see [Pa]).

35. Define the set E of expressions by (i) Con is an expression; (ii) if A, B are expressions so are $(\sim A)$, $(A \& B)$, and $\text{Con}(A)$. Each expression ϕ in E determines a sentence ϕ^* in PA by taking Con^* to be " PA is consistent," $(\sim A)^*$ to be $\sim(A^*)$, $(A \& B)^*$ to be $A^* \& B^*$, and $\text{Con}(A)$ to be " $PA + 'A^*'$ is consistent." *The set of expressions $\phi \in E$ such that ϕ^* is true is recursive.*

The formalized second incompleteness theorem reads $\sim \text{Con}(\text{Con} \& \sim \text{Con}((\sim \text{Con})))$.

36. *There is a consistency proof of the classical system of second-order arithmetic, $\Sigma_2^1 - AC$, within the intuitionistic theory $\text{ID}(\epsilon_0)$ of ϵ_0 iterated inductive definitions of species of natural numbers (Feferman).*

There is a consistency proof of classical $\Sigma_2^1 - AC$ within classical $\text{ID}(\epsilon_0)$ (see

[Fr 3]) and [Fe 2]). Intuitionistic ID(1) and classical ID(1) have the same proof-theoretic strength, but it is not known whether intuitionistic ID(2) and classical ID(2) have the same proof-theoretic strength (see [Zu]). There is a consistency proof of classical Π_1^1 - CA within intuitionistic ID(ω) (see [Fe 2] and [Zu]).

37. *The set of propositional formulae which, when added as a scheme to intuitionistic first-order arithmetic with Church's Thesis, results in a consistent extension of intuitionistic first-order arithmetic, is decidable.*

It is commonly known that $p \vee \sim p$ is not such a formula. The formula $(\sim p \rightarrow (q \vee r)) \rightarrow ((\sim p \rightarrow q) \vee (\sim p \rightarrow r))$ is such a formula.

38. *HA^ω with extensionality and the full axiom of choice is a conservative extension of HA .*

HA^ω with the full axiom of choice is a conservative extension of HA (see [Goo 1], [Goo 2]).

Propositional calculi.

39. *Every finitely axiomatized theory (allowing the rule of substitution) in intuitionistic propositional calculus is decidable (Harrop).*

This statement is true in the intuitionistic propositional calculus based only on the connectives $\rightarrow$, $\&$, $\sim$ (see [Mc]).

40. *There is a decision procedure for determining whether a figure A/B represents a valid rule of inference in the intuitionistic propositional calculus (where A, B are formulae in the propositional calculus).*

41. *There is a set of formulae T in the propositional calculus based on $\perp$, $\&$, $\vee$, $\rightarrow$ obeying (i) $\perp \notin T$, (ii) $A \& B \in T$ if and only if $A, B \in T$, (iii) $A \vee B \in T$ if and only if $A \in T$ or $B \in T$, and (iv) $A \rightarrow B \in T$ if and only if every substitution that puts A in T also puts B in T . Furthermore this T is not unique.*

We have observed that such a T must properly contain all provable formulae of the intuitionistic propositional calculus.

42. The V -calculus is propositional calculus based on $\sim$, $\&$, and the operator V . Thus $\sim A$, $A \& B$, $V(A)$ are V -formulae if A, B are V -formulae. There is a set T of V -formulae such that (i) each atom is in T , (ii) $\sim A \in T$ if and only if $A \notin T$, (iii) $A \& B \in T$ if and only if $A, B \in T$ and (iv) $V(A) \in T$ if and only if every substitution instance of A (by V -formulae) is in T . Furthermore this T is not unique.

If, in clause (iv), "(by V -formulae)" is replaced by "by propositional formulae," then the clauses constitute a noncircular inductive definition of a set S of V -formulae, and we have observed that this S does not obey the original clause (iv).

Set theory.

43. *Quine's "New Foundations" is consistent (Quine).*

Quine's "New Foundations" without extensionality is consistent (see [Je]).

44. *Every Π_2^0 sentence provable in classical Zermelo set theory is provable in intuitionistic Zermelo set theory (?).*

This statement is true for Π_1^0 by adapting [Fr 8].

45. *Any Π_2^0 sentence provable in classical ZF is provable in the system ZFI of [My].*

Any Π_1^0 sentence provable in classical ZF is provable in the system ZF^- of [Fr 8] (by adapting [Fr 8]). ZF^- is the same as ZFI except that the collection axiom scheme

is used in ZF^- instead of the replacement axiom scheme in ZFI . Hence ZFI is a subsystem of ZF^- . It is not known whether ZF^- proves the consistency of ZFI .

46. *The system ZFI of [My] does not prove the collection axiom scheme of the system ZF^- of [Fr 8].*

47. *The existential definability property fails for the system ZF^- of [Fr 8].*

ZFI obeys the existential definability property and the disjunction property (see [My]). We have shown that ZF^- of [Fr 8] obeys the disjunction property.

RECURSION THEORY

Closure ordinals.

48. *If $\omega_1 < \alpha < \omega_2$ is admissible, $\text{cf}(\alpha) = \omega_1$, then α is the first ordinal admissible in some $x \subset \omega_1$ (?).*

It is commonly known that the first uncountable ordinal admissible in any $x \subset \omega_1$ has cofinality ω_1 . Jensen has shown the above statement to be true under the axiom of constructibility. If $\alpha < \omega_1$ is admissible then α is the first ordinal admissible in some $x \subset \omega$ (see [Sa 4] and [FJ]).

49. *If $\omega_1 < \alpha < \omega_2$ is stable, $\text{cf}(\alpha) = \omega_1$, and ω_2 is inaccessible in L , then α is the first ordinal stable in some $x \subset \omega_1$.*

It is commonly known that the first uncountable ordinal stable in any $x \subset \omega_1$ has cofinality ω_1 .

50. *If $\alpha < \omega_1$ is stable and $\mathcal{P}(\omega) \not\subseteq L$, then α is the first ordinal stable in some $x \subset \omega$.*

If ω_1 is inaccessible in L , then every stable $\alpha < \omega_1$ is the first ordinal stable in some $x \subset \omega$ (see [Fr 13]).

51. *If $\alpha < \omega_1$ and $L_\alpha \models ZF$, then α is the first ordinal such that $L_\alpha(x) \models ZF$, for some $x \subset \omega$ (Sacks).*

This statement is true if the power set axiom is deleted from ZF (see [Sa 2]).

Admissible sets.

52. *There is an admissible set A such that any two disjoint A -r.e. sets can be separated by an A -recursive set (Simpson).*

We have observed that for admissible sets A , the reduction principle for A -r.e. sets implies the existence of a partial A -recursive enumeration of the partial $\{0, 1\}$ -valued A -recursive functions, which in turn implies the existence of two disjoint A -r.e. sets which cannot be separated by an A -recursive set.

53. *In the lattice of the countable admissible sets under inclusion, every element has a minimal upper bound.*

This statement is true for countable transitive sets satisfying ZFC (see [Sa 3]).

α -degrees.

54. *For each admissible ordinal α there is a minimal α -degree (MacIntyre).*

For each Σ_2 -admissible ordinal α , there is a minimal α -degree (see [Sho]). For each countable admissible ordinal α , there is a minimal α -degree (see [Ma]).

55. *For each admissible ordinal α there is a minimal pair of α -r.e. degrees (Lerman, Sacks).*

For each admissible ordinal α , there is an incomparable pair of α -r.e. degrees (see [SS]). For each projectible admissible ordinal α there is a minimal pair of α -r.e. degrees (see [LS]).

56. *There is an admissible ordinal $\alpha < \omega_1$ such that the first-order theory of the α -degrees, under $<_\alpha$, is the same as the first-order theory of the Turing degrees, under $<_T$ (?).*

It is open whether this statement is true for all admissible $\alpha < \omega_1$.

Hyperdegrees.

57. *Every recursive tree of finite sequences of natural numbers with uncountably many infinite paths has infinite paths of each hyperdegree $\geq$ Kleene's $\mathcal{O}$.*

This statement is true if the recursive tree has no infinite hyperarithmetic paths (see [Fr 10]). This statement is true if "each hyperdegree $\geq$ Kleene's $\mathcal{O}$ " is replaced by "hyperdegree Kleene's $\mathcal{O}$ " (see [Har]). For every nonrecursive f there is an infinite recursive 0, 1-tree with no infinite recursive paths such that f is not recursive in any infinite path (see [Jo S2]).

58. *Every uncountable Σ_1^1 subset of ω^ω contains two elements whose join has any prescribed hyperdegree $\geq$ Kleene's $\mathcal{O}$ (Friedman, Simpson).*

It is commonly known that this statement is true if the set contains an element of hyperdegree $\geq$ Kleene's $\mathcal{O}$. This statement is open if " $\geq$ " is eliminated. There is an uncountable Σ_1^1 subset of ω^ω every element of which has minimal hyperdegree (see [Simp]).

59. *There is a sequence of hyperdegrees with no minimal upper bound (Sacks).*

Every sequence of Turing degrees has a minimal upper bound (see [Sa 1]). The hyperdegrees appearing in any $A \cap \omega^\omega$, for countable admissible sets A satisfying $\Sigma_1 - DC$, have a minimal upper bound (see [Sa 4]). It is open whether there is a sequence of hyperdegrees below Kleene's $\mathcal{O}$, with no minimal upper bound.

Recursively enumerable sets.

60. *For every nonrecursive r.e. $A \subset \omega$ and every r.e. degree d with $d' = 0'$, there is a $B \in d$ and an automorphism of the lattice of r.e. sets under inclusion which sends A to B (Soare).*

For every nonrecursive r.e. $A \subset \omega$ there is an r.e. $B \in 0'$ and an automorphism of the lattice which sends A to B (see [So 2]). It is not known whether for every r.e. $A \in 0'$ there is an r.e. $B \notin 0'$ and an automorphism of the lattice which sends A to B . For any two maximal r.e. A, B there is an automorphism of the lattice which sends A to B (see [So 1]).

61. *The first-order theory of the lattice of r.e. sets under inclusion is decidable (Lachlan).*

The Π_2 theory of the lattice of r.e. sets under inclusion is decidable (see [La 1]).

Singletons.

62. *There is a Π_1^0 singleton that is not of the same arithmetic degree as any Kleene H -set.*

This statement is commonly known to be false for Turing degree. It is open for arithmetic singletons.

63. *Every element of every countable arithmetic subset of ω^ω is an arithmetic singleton* (McLaughlin).

Every nonempty countable arithmetic set possesses an arithmetic singleton (see [Ta]). Every element of every countable Σ_1^1 set is hyperarithmetical (see [Har]). Not every hyperarithmetical set of numbers is an arithmetic singleton (see [Fe 1]).

Theories of the hyperarithmetical sets.

64. *There is an ω -model of $\Delta_1^1 - CA$ which is not a model of $\Sigma_1^1 - AC$* (Kreisel).

There is an ω -model of $\Sigma_1^1 - AC$ which is not a model of $\Sigma_1^1 - DC$ (see [Fr 1]). It is open whether $\Sigma_1^1 - AC$ can be derived from $\Delta_1^1 - CA$, $\Sigma_1^1 - DC$, $\Sigma_1^1 - AC$, and $\Delta_1^1 - CA$ have the same provable Π_2^1 sentences (see [Fr 1]).

65. *There is an ω -model of $\Sigma_1^1 - DC$ which does not satisfy "every Δ_1^1 set is hyperarithmetical"* (Kreisel).

The hyperarithmetical sets form the minimum ω -model of $\Delta_1^1 - CA$, $\Sigma_1^1 - AC$, and $\Sigma_1^1 - DC$ (see [Kr]) and satisfy "every Δ_1^1 set is hyperarithmetical" (see [K1 2]).

Turing degrees.

66. *The Turing degree 0^ω is minimal over some Turing degree* (Jockusch, Soare).

Every sufficiently large Turing degree is minimal over some Turing degree (see [Jo]). No 0^n is minimal over any Turing degree (see [Jo SI]).

67. *There is an initial segment of Turing degrees of length ω_1* (Yates).

For each $\alpha < \omega_1$ there is an initial segment of Turing degrees of length α (see [Hu]). This statement is consistent with ZFC (see [Ya]).

68. *The Turing degrees at or above any given Turing degree, under $<_T$, are isomorphic to all the Turing degrees, under $<_T$* (Rogers).

This is open if "any" is replaced by "some", and/or "isomorphic" is replaced by "elementary equivalent". Simpson has shown that the relation " d_1 is constructible from d_2 " is first-order definable in the structure of the Turing degrees, under $<_T$ and the jump operator. Hence there is a model of ZFC + "there is a Turing degree such that the Turing degrees at or above it, under $<_T$ and jump, are not elementary equivalent to all the Turing degrees, under $<_T$ and jump".

69. *There is an e such that for all $f \in \omega^\omega$ we have $f <_T W_e^f <_T f'$, and if $f =_T g$ then $W_e^f =_T W_e^g$* (Sacks).

The usual construction provides an e without the second condition.

70. *The first-order theory of the Turing degrees under $<_T$ is not hyperarithmetical* (Sacks).

This theory is not recursively enumerable (see [La 2]). It is open whether this theory is arithmetical, or constructible, or analytic.

Simpson's result cited in 68 above implies that there is a model of ZFC + "the first-order theory of the Turing degrees under $<_T$ and jump, is nonconstructible."

71. *There is a Π_1^1 path through Kleene's $\mathcal{O}$ such that the only hyperarithmetical sets recursive in it are recursive* (?).

It is not known whether Kleene's $\mathcal{O}$ is recursive in every Π_1^1 path through Kleene's $\mathcal{O}$. There is a Π_1^1 path through Kleene's $\mathcal{O}$ in which Kleene's $\mathcal{O}$ is recursive, and every initial segment is recursive (see [Fr 15]).

SET THEORY

Combinatorial.

72. *If κ is a measurable cardinal then every subset of κ contains or is disjoint from a closed subset of κ of type ω_1 .*

Silver, Solovay and Prikry have shown that if $V = L$ there are arbitrarily long sets of ordinals which neither contain nor are disjoint from a closed set of ordinals of type ω_1 with smaller supremum (see [Pr 2]). Every stationary subset of ω_1 contains arbitrarily long countable closed subsets (see [Fr 12]). It is open whether every subset of ω_2 contains or is disjoint from a closed subset of ω_2 of type ω_1 .

73. *If $ZF + \text{"dependent choice"} + \text{"every subset of } \omega_1 \text{ contains or is disjoint from a closed unbounded subset of } \omega_1\text{"}$ is consistent, so is $ZFC + \text{"there are two measurable cardinals"}$ (?).*

It is commonly known that the statement is true if "two" is deleted. Kunen has observed that if $ZF + \text{"dependent choice"} + \text{"every constructible subset of } \omega_1 \text{ contains or is disjoint from a closed unbounded subset of } \omega_1\text{"}$ is consistent, so is $ZFC + \text{"}0^\# \text{ exists"}$.

74. *If every Δ_2^1 set of countable ordinals contains or is disjoint from a closed unbounded set of countable ordinals, then $0^\#$ exists (Kunen).*

It is commonly known that every Π_1^1 set of countable ordinals contains or is disjoint from a closed unbounded subset of ω_1 .

c-degrees.

75. *If $\omega_1^L < \omega_1$ then some increasing sequence of degrees of constructibility has no minimal upper bound.*

Every sequence of Δ_2^1 -degrees has a minimal upper bound (see [Fr 13]).

76. *There are Π_2^1 singletons x such that $0 <_c x <_c 0^\#$, which are of incomparable c-degree (Solovay).*

The c -degree of every Π_2^1 singleton is comparable with $0^\#$ (see [HK]). It is open whether there are Π_2^1 singletons x such that $0 <_c x <_c 0^\#$. It is open whether the c -degrees of Π_2^1 singletons are well-ordered under $<_c$.

Determinateness.

77. *$ZFC + \text{"determinateness for sets in } OD(\mathbf{R})\text{"}$ does not imply the continuum hypothesis, provided the former is consistent (?).*

Kunen and Moschovakis have shown that if determinateness holds for sets in $L(\mathbf{R})$ then there is a model of $ZF + \text{projective determinateness} + \text{the negation of the continuum hypothesis}$. Solovay has shown that $ZFC + \text{"determinateness for sets in } OD(\mathbf{R})\text{"}$ is consistent with the continuum hypothesis if the former is consistent.

78. *$ZFC + \text{"every set of Turing degrees in } OD(\mathbf{R}) \text{ contains or is disjoint from a cone of Turing degrees"}$ does not imply determinateness for sets in $OD(\mathbf{R})$ (Martin).*

Determinateness for sets in $OD(\mathbf{R})$ implies that every set of Turing degrees in $OD(\mathbf{R})$ contains or is disjoint from a cone of Turing degrees (see [Mar 1]). The statement is open if " $OD(\mathbf{R})$ " is replaced by " Δ_2^1 " in the first and/or second instance.

79. *Every countable transitive model of $ZFC +$ projective determinateness has a Boolean extension in which projective determinateness fails* (Solovay).

This is commonly known to be false for $L(\mathcal{R})$ determinateness. It is open for $OD(\mathcal{R})$ determinateness, and for Δ_2^1 determinateness.

80. *$ZFC +$ “determinateness for Σ_2^1 sets” does not imply the existence of $0^\#$* (Martin).

There is a Σ_1^1 set which is not determined in any Boolean extension of L (see [Fr 5]). If $0^\#$ exists then determinateness for Σ_1^1 sets holds (see [Mar 2]). Determinateness for a class of sets a bit beyond Σ_1^1 is equivalent to the existence of $0^\#$ (see [Mar 3]).

81. *If every Σ_1^1 set of Turing degrees contains or is disjoint from a cone of Turing degrees, then $\omega_1^{L(x)} < \omega_1$ for all $x \subset \omega$.*

Simpson has shown that there is a Σ_1^1 set of Turing degrees which is satisfied to neither contain nor be disjoint from any cone of Turing degrees in any Boolean extension of L (see [Simp]).

82. *Determinateness for Borel sets is not provable in ZFC (?)*.

Determinateness for Borel sets is not provable in Zermelo set theory (see [Fr 6]). Determinateness holds for $F_{\sigma\delta\sigma}$ sets (see [Par]). If certain Ramsey type cardinals exist, then determinateness holds for Borel sets (see [Mar 2]). For some recent refinements, see [Mar 5].

Forcing and independence.

83. *The well-known singular cardinal problems (?)*.

84. *There is a model of $ZFC +$ “to each ω -closed partial ordering $\mathcal{P}$ in which every set of power ω_2 contains comparable elements, and to each set F of $\mathcal{P}$ -dense sets of power $\leq \omega_2$, there is a set of mutually comparable elements of $\mathcal{P}$ which meet every element of F ” (?)*.

This is true if “ ω -closed” is deleted and “ ω_2 ” is replaced by “ ω_1 ” (see [MS]).

85. *There is a model of $ZFC + 2^\omega > \omega_1 +$ “any two suborderings of the real line whose intersections with every open interval of the line has cardinality ω_2 are order isomorphic” (?)*.

This is true if “ ω_2 ” is replaced by “ ω_1 ” (see [Bau]).

86. *There is a model of $ZFC +$ “there is a conconstructible real” + “the constructible reals have a perfect subset”* (Prikry).

This is open if “have a perfect subset” is replaced by “is an uncountable Borel set,” or “is an uncountable Σ_1^1 set”. This is true for “is an uncountable Π_1^1 set” (see [MS]).

87. *For each $2 < n \leq \infty$ there is a model of $ZFC +$ “the constructible sets of integers are precisely the Δ_n^1 sets of integers”*.

This is open for any $2 < n \leq \infty$.

88. *There is a model of $ZFC +$ “there exists a measurable cardinal” + “every $\alpha < \omega_2$ is the length of some Δ_2^1 well-ordering of ω^ω ”* (Solovay).

This is true if “there exists a measurable cardinal” is eliminated (see [MS]), and open if it is replaced by “ $x^\#$ exists for all $x \subset \omega$ ”. Every well-founded Σ_2^1 relation has ordinal $< \omega_2$ (see [Ma 4]).

89. *Let α be least such that $L_\alpha \models ZFC$. Then there is an $x \subset \omega$ such that $L_\alpha(x) \models ZFC$,*

and such that the cardinals of $L_\alpha(x)$ are not eventually the same as the cardinals of L_α (Jensen, Solovay).

90. *There is a model of ZF + “every ordinal has cofinality ω ” (Levy).*

There is a model of ZF + “ ω_1 has cofinality ω ” (see [Co]).

91. *Every countable transitive model of ZFC has an extension with the same cardinals satisfying ZFC + $(\exists x \subset \omega)(\omega_1^{L(x)} = \omega_1)$.*

In certain cases, such extensions are constructed in [JS]. It is commonly known that no model M of ZFC + “there is a proper class of measurable cardinals” has such a *Boolean* extension (via a notion of forcing $\mathcal{P} \in M$).

92. *There is a countable transitive model M of ZFC + GCH and an $x \subset \omega$ such that $M[x]$ is a cardinal preserving Boolean extension of M satisfying $2^\omega > \omega_1$ (Jensen, Solovay).*

This is open if “cardinal preserving Boolean extension of M ” is replaced by “model of ZFC”. It is commonly known that if M is a countable transitive model of ZFC + GCH, $x \subset \omega$, and either $M \models (\exists x \subset \omega) (V = L(x))$ or $M[x]$ is a forcing extension of M via a c.c.c. partial ordering, then $M[a] \models$ GCH.

Measurability.

93. *Every Π_3^1 sentence provable in ZFC + “all sets of reals in $\text{OD}(\mathcal{R})$ are Lebesgue measurable” is already provable in ZFC (Solovay).*

This statement is open if “ Π_3^1 ” is replaced by “ Π_1^0 ” and/or “is Lebesgue measurable” is replaced by “has the property of Baire” or “contains or is disjoint from a perfect set”, and/or “ $\text{OD}(\mathcal{R})$ ” is replaced by “ Δ_3^1 ”. Every Π_3^1 sentence provable in ZFC + “all OD sets of reals are Lebesgue measurable and have the property of Baire”, or in ZFC + “all PCA sets are Lebesgue measurable and have the property of Baire” is provable in ZFC (by adapting [Sol] and [MS] respectively).

94. *Some of the usual models (e.g. in [ST]) constructed to satisfy Martin’s axiom also satisfy “all sets of reals in $\text{OD}(\mathcal{R})$ are Lebesgue measurable” (Solovay).*

Martin’s axiom + $2^\omega > \omega_1$ implies that every PCA set is Lebesgue measurable and has the property of Baire (see [MS]). The statement is open if “Some” is replaced by “All” and/or “ $\text{OD}(\mathcal{R})$ ” is replaced by “ Δ_3^1 ” and/or “is Lebesgue measurable” is replaced by “has the property of Baire” or “contains or is disjoint from a perfect set”. It is commonly known that the models of [ST] are not all elementary equivalent.

Models of set theory.

95. *There is a transitive model of ZFC + $\mathcal{P}(\omega) \not\subseteq L$ with height that of the minimum model M of ZFC, which contains a real not present in any generic extension of M (?).*

96. *Let M be a countable transitive model of ZFC + $V \neq L + A$. Then there is a transitive model N of ZFC + $V \neq L + A$ such that $M \cap \text{On} = N \cap \text{On}$ and $M \neq N$ (Tharp).*

This is true provided $M \cap \text{On}$ is a member of a certain technically defined closed unbounded subset of ω_1 , or if M satisfies “ $0^\#$ exists”, or if M satisfies “ $V \neq \text{HOD}$ ” (see [Fr 9]).

97. *Every transitive model M of ZFC such that $M \cap \text{On}$ is not a cardinal is properly included in a transitive model of ZFC with the same ordinals.*

It is commonly known that this statement is true if $V = L$, and the cardinal of $M \cap \text{On}$ is regular. It is open whether the full statement follows from $V = L$.

98. *There is a model M of ZFC with (a) nonstandard integers, (b) $\{a \in \text{dom}(M): M \models a \in \omega\}$ has power ω , (c) $\{a \in \text{dom}(M): M \models a \in \omega_1\}$ has power ω_1 , and (d) $\{a \in \text{dom}(M): M \models a \in \omega_2\}$ has power ω_2 .*

Burgess has observed that the transfer propositions $\omega_2, \omega \rightarrow \omega_3, \omega_1$ and $\omega_3, \omega_1 \rightarrow \omega_2, \omega$ together imply this statement. Jensen has proved that these transfer propositions hold if $V = L$.

99. *There is a transitive model M of ZF such that $M \cap \text{On} < \omega_1$ and $M \cap \omega^\omega$ has outer measure 1 (Keisler).*

It is commonly known that this is true if Martin's axiom holds.

100. *There exists a transitive model M of ZF with countably many ordinals, which is not properly included in any transitive model of ZF with the same ordinals.*

Every countable transitive model $M \models \text{ZFC}$ with $M \cap \text{On} = \alpha$ can be extended to a model of ZF of power $\beth_\alpha$ with the same ordinals (see [Fr 14]).

Ramsey cardinals.

101. *Let κ be a Ramsey cardinal. The intersection of all transitive classes containing On satisfying $\text{ZFC} + "$ κ is Ramsey", satisfies $\text{ZFC} + "$ κ is Ramsey" + $\text{GCH} + "$ there exists a strong Σ_3^1 well-ordering of the reals" (Silver).*

This is true if "Ramsey" is replaced by "measurable" (see [Si 2]). None of the four conclusions are known to hold.

102. *Every countable model of $\text{ZFC} + "$ κ is a Ramsey cardinal" has cardinal preserving generic extensions with the same $V(\kappa)$, but in which κ is satisfied to have cofinality ω (Prikry).*

This is true if "Ramsey" is replaced by "measurable" (see [Pr 1]). Magidor has shown how to change the cofinality of cardinals obeying an augmented measurability condition to various cardinals $> \omega$ (see [Mag 2]), and Mitchell has shown that the cofinality of a Ramsey cardinal cannot always be changed to a smaller measurable cardinal.

POSTSCRIPT (ADDED SEPTEMBER 1974).

We report the following recent progress on our problem list.

With regard to problem 56, Shore has shown that, for each Σ_2 admissible ordinal α , there is a minimal pair of α -r.e. degrees.

Problem 57 has been solved affirmatively by D. A. Martin and the author, independently. Modify problem 57 as follows.

57*. *Every recursive tree of finite sequences of natural numbers with uncountably many infinite paths has infinite paths of each λ -degree $\geq$ Kleene's $\mathcal{O}$, for some recursive ordinal λ .*

Harrington and Kechris (*A basis for Σ_3^0 sets of reals*, to appear) have shown that every Σ_3^0 sets of reals which contains reals of arbitrarily high λ -degree in the hyperarithmetical hierarchy contains reals of every λ -degree $\geq$ Kleene's $\mathcal{O}$ (where λ is recursive).

Problem 58 has been solved affirmatively by L. Harrington. Modify problem 58 as follows.

58*. *There is a model of ZFC in which there is an $x \subset \omega$ such that the countable ordinals admissible in x are exactly the countable recursively inaccessible ordinals.*

With regard to problem 62, J. Steel has constructed f, g such that g is a Π_1^0 singleton relative to f , yet (f, g) is not of the same arithmetic degree as any H -set relative to f .

With regard to problem 63, J. Steel has constructed an f and a countable set arithmetic relative to f , which has a member which is not an arithmetic singleton relative to f .

With regard to problem 70, S. Simpson has shown that the truth set of $(\omega, ', +, \cdot, \mathcal{P}(\omega), \epsilon)$ is recursively isomorphic to the elementary theory of the Turing degrees under $<_T$ and Turing jump.

Problem 74 has been solved negatively by L. Harrington. In fact he proves that the statement is false even if " Δ_2^1 " is replaced by "projective."

With regard to problem 83, Silver has shown that if the GCH holds below a singular cardinal of uncountable cofinality, it holds at that cardinal. Jensen has shown that if every bounded subset of a singular cardinal of uncountable cofinality is constructible, so is every subset. It is not known whether either of these results hold for all singular cardinals.

Problem 87 has been solved affirmatively by L. Harrington. Modify problem 87 as follows.

87*. *There is a model of ZFC + "every set of integers constructible in $x \subset \omega$ is Δ_3^1 in x ".*

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